

int left = 0, right = mat[0].length - 1;

int top = 0, bottom = mat.length - 1;

while (top <= bottom && left <= right) {

for (int i = left; i <= right; i++) {

System.out.print(mat[top][i]);

top++;

for (int i = top; i <= bottom; i++) {

System.out.print(mat[i][right]);

right--;

if (top <= bottom) {

for (int i = right; i >= left; i--) {

System.out.print(mat[bottom][i]);

bottom--;

if (left <= right) {

for (int i = bottom; i >= top; i--) {

System.out.print(mat[i][left]);

left++;

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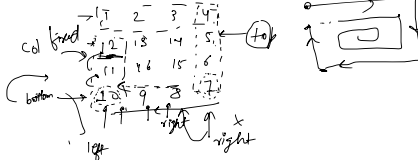
}

}

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}



1 2 3 4 5 6

7 8 9 10

11 12 13 14

15 16

17 18 19 20

21 22 23 24

25 26 27 28

29 30 31 32

33 34 35 36

37 38 39 40

41 42 43 44

45 46 47 48

49 50 51 52

53 54 55 56

57 58 59 60

61 62 63 64

65 66 67 68

69 70 71 72

73 74 75 76

77 78 79 80

81 82 83 84

85 86 87 88

89 90 91 92

93 94 95 96

97 98 99 100

101 102 103 104

105 106 107 108

109 110 111 112

113 114 115 116

117 118 119 120

121 122 123 124

125 126 127 128

129 130 131 132

133 134 135 136

137 138 139 140

141 142 143 144

145 146 147 148

149 150 151 152

153 154 155 156

157 158 159 160

161 162 163 164

165 166 167 168

169 170 171 172

173 174 175 176

177 178 179 180

181 182 183 184

185 186 187 188

189 190 191 192

193 194 195 196

197 198 199 200

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16

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1 2 3 4 4 3 2 1

1 2 3 4 4 3 2 1

1 2 3 4 4 3 2 1

1 2 3 4 4 3 2 1

1 2 3 4 4 3 2 1

1 2 3 4 4 3 2 1

1 2 3 4 4 3 2 1

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1 2 3 4 4 3 2 1

1 2 3 4 4 3 2 1

1 2 3 4 4 3 2 1

1 2 3 4 4 3 2 1

1 2 3 6 9 8 7 4 5

```

public class Array_2d_second {
    public static void main(String[] args) {
        int [][] mat = {{1,2,3},{4,5,6},{7,8,9}};
        int left=0, right=mat[0].length-1;
        int top=0, bottom=mat.length-1;
        while (top <= bottom && left <= right) {
            for (int i = left; i <= right; i++) {
                System.out.print(mat[top][i]);
            }
            top++;
            for (int i = top; i <= bottom; i++) {
                System.out.print(mat[i][right]);
            }
            right--;
            if (top <= bottom) {
                for (int i = right; i >= left; i--) {
                    System.out.print(mat[bottom][i]);
                }
                bottom--;
            }
            if (left <= right) {
                for (int i = bottom; i >= top; i--) {
                    System.out.print(mat[i][left]);
                }
                left++;
            }
        }
    }
}

```

* matrix multiplication

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$$

$$B = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$$

$$C = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 3 & 6 & 9 \end{bmatrix}_{3 \times 3}$$

imp rule $\rightarrow$ first matrix's col = 2nd matrix's row

$$\begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 3 & 6 & 9 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 3 & 6 & 9 \end{bmatrix}$$

$$\begin{bmatrix} 2 & 4 & 6 \\ 3 & 6 & 9 \end{bmatrix}_{3 \times 3}$$

1) first matrix's col = 2nd matrix's row

2) result matrix's dimension = first matrix row $\times$ 2nd matrix col.

$$\begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 3 & 6 & 9 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 3 & 6 & 9 \end{bmatrix}$$

$$a = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 3 & 6 & 9 \end{bmatrix}_{3 \times 3}, b = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 3 & 6 & 9 \end{bmatrix}_{3 \times 3}$$

$$C = \begin{bmatrix} 2 \times 1 + 1 \times 1 & 2 \times 1 + 1 \times 1 \\ 1 \times 1 + 2 \times 1 & 1 \times 1 + 2 \times 1 \end{bmatrix}_{2 \times 2} \Rightarrow \begin{bmatrix} 3 & 3 \\ 3 & 3 \end{bmatrix}_{2 \times 2}$$

$$a = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 3 & 6 & 9 \end{bmatrix}_{3 \times 3}, b = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 3 & 6 & 9 \end{bmatrix}_{3 \times 3}$$

$$\begin{bmatrix} 1 \times 3 + 2 \times 2 + 3 \times 1 \end{bmatrix}_{1 \times 1}$$

$$= \begin{bmatrix} 10 \end{bmatrix}_{1 \times 1}$$

Code:-

int row1 = a.length;

int col1 = a[0].length;

int row2 = b.length;

int col2 = b[0].length;

int[][] C = new int[row1][col2];

if (col1 == row2) {

for (int i = 0; i < row1; i++) {

for (int j = 0; j < col2; j++) {

for (int k = 0; k < col1; k++) {

C[i][j] += a[i][k] * b[k][j];

}

else {

System.out.println("the matrices cannot be multiplied");

}

col = a[0].length;

$$\begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 3 & 6 & 9 \end{bmatrix}$$

$$(1 \times 1) + (1 \times 1) + (1 \times 1)$$

1st row

is traverse on the rows.

not traverse on col.

$$1 \times 1 + 1 \times 1 + 1 \times 1$$

$$= \begin{bmatrix} 3 & 3 & 3 \\ 3 & 3 & 3 \\ 3 & 3 & 3 \end{bmatrix}$$

* transpose of matrix